

Princeton University
Intermediate Micro: Math Approach (ECO 310)
Spring 2025 Midterm—Andrea Wilson

Instructions: This is intended to be an 80-minute exam, and you have 100 minutes altogether (10 minutes for uploading your answers and 10 minutes of cushion). There are 6 multiple choice questions, each worth 4 points, which you can answer directly on Canvas. And three short answer questions, worth a total of 86. In the Short Answer section, REMEMBER TO SHOW YOUR WORK!, YOU'LL LOSE SIGNIFICANT CREDIT FOR ANSWERS THAT APPEAR BY MAGIC.

- please include the signed honor pledge in your short answer submission
- the actual total is 110, but I'll grade out of 100. So you can costlessly skip a 10-point question.

Multiple Choice Questions

1. A consumer's preferences over goods 1 and 2 are represented by $u(x_1, x_2) = x_1^{0.2}x_2^{0.4}$. Good 1 costs \$1 per unit, good 2 costs \$2 per unit, and income is \$15. How many units of good 2 should this consumer buy?
 - (a) 3
 - (b) 4
 - (c) 5
 - (d) None of these is correct
2. On Monday, Homer Simpson buys a bundle of food and drink that costs \$90. On Tuesday, he buys a different bundle, for \$100, that would have cost \$80 at the Monday prices. In which of the following situations does he violate WARP:
 - (a) If the Monday bundle would cost \$100 at the Tuesday prices.
 - (b) If the Monday bundle would cost more than \$100 at the Tuesday prices.
 - (c) Homer does not violate WARP.
3. Assume utility $u(x_1, x_2) = \min\{4x_1, x_2\}$. The Hicksian demand bundle is given by:
 - (a) $(4u^*, u^*)$
 - (b) $(\frac{u^*}{4}, u^*)$
 - (c) $(u^*, \frac{u^*}{4})$
 - (d) $(u^*, 4u^*)$
4. There are two assets available: a risk-free asset, with a 10% return rate, and a risky asset, with a 20% expected return rate and variance $1/4$. If an individual divides his money between the two assets so that the standard deviation on his overall investment is $1/4$, then his overall expected return rate is:
 - (a) .15
 - (b) .1

- (c) .2
(d) None of these is correct
5. Jessica has utility $u(x_1, x_2) = x_1 + 2x_2$. She sets up a Lagrangian to find her best affordable bundle with income \$m. In which of the following cases do you expect the non negativity constraint on good 1 to be NON-binding?
- (a) $p_1 = 2, p_2 = 5$
(b) $p_1 = 2, p_2 = 1$
(c) $p_1 = 2, p_2 = 3$
(d) More than one of these is correct.
6. Melsi has utility $u(B, T) = BT$ from B beef sticks and T toys. Initially, Melsi buys 16 beef sticks and 4 toys, when each beef stick costs \$1 and each toy costs \$4. What is the change in beef sticks due to substitution effects if the beef stick price rises to \$4? (no change in toy price).
- (a) -16
(b) -8
(c) -4
(d) None of these is correct

Short Answer Questions

1. (32) Demand

Melsi's utility over beef sticks (B) and toys (T) is

$$u(B, T) = \left(B^{\frac{s-1}{s}} + T^{\frac{s-1}{s}} \right)^{\frac{s}{s-1}}$$

where s is a nonnegative parameter.

- (a) (10) Find Melsi's regular (Marshallian) demand functions, with p_B and p_T the beef stick and toy prices.

You should get

$$(B, T) = \left(\frac{m}{p_B^s (p_B^{1-s} + p_T^{1-s})}, \frac{m}{p_T^s (p_B^{1-s} + p_T^{1-s})} \right)$$

- (b) (12) Show that for two of the options below (they're all true but pick the two you want to show), your demand functions in (a) approach those for:

- perfect complements as $s \rightarrow 0$
- perfect substitutes as $s \rightarrow \infty$
- Cobb-Douglas (with equal exponents) as $s \rightarrow 1$

- (c) (10) Take as given that Melsi's expenditure function is $\frac{p_B u^*}{\left(1 + \left(\frac{p_B}{p_T}\right)^{s-1}\right)^{\frac{1}{s-1}}}$. Use this to

figure out how much money she would need to get utility $u^* = 1$ in the limit as $s \rightarrow 0$. Does your answer match your intuition for perfect complements? (Either explain why it does, or why you think there might be an error).

2. (28) Expected Utility: Risk Aversion

(a) Consider the following bet:

$$\begin{cases} \text{win \$200} & \text{chance } \frac{1}{3} \\ \text{lose \$100} & \text{chance } \frac{2}{3} \end{cases}$$

- i. (5) Explain (intuitively is fine) why a risk-averse decision-maker (DM) would reject this at ANY wealth level
- ii. (12) Next, explain why a risk-averse DM would find playing it twice even worse than playing it once. Specifically, complete the following proof:
 - we know (1): $u(w_0 + 200) > \frac{1}{3}u(w_0 + 400) + \frac{2}{3}u(w_0 + 100)$ (why?)
 - we know (2): $u(w_0 - 100) > \frac{1}{3}u(w_0 + 100) + \frac{2}{3}u(w_0 - 200)$ (why?)
 - Now take a weighted sum of these inequalities ($\alpha(1) + (1 - \alpha)(2)$), and show that for suitably chosen α , you have answered the question
- (b) (11) Now modify the bet in (a) to have the same prizes, but now equally likely. And assume $u(w) = \ln w$. Could it ever happen that a poor decision-maker accepts the bet, while a wealthier one does not? (Either find initial wealth levels where this happens or show that it's impossible).

3. (26) Expected Utility: Optimal Decisions

A risk-averse decision-maker with wealth w_0 would like to invest \$100 in total, some in “asset 1” and some in “asset 2”. The two assets are correlated in the following way (note, you can do this even if you have no idea what correlation is):

- in state A (chance $\frac{1}{3}$), return rates are $r_1^A > r_2^A$ for assets 1 and 2
- in state B (chance $\frac{1}{3}$), both assets have the same return rate, r^B
- in state C (chance $\frac{1}{3}$), return rates are $r_1^C < r_2^C$ for assets 1 and 2

- (a) (8) Write out the expected utility from investing $\$x$ in asset 1 and $\$100 - x$ in asset 2.
- (b) (12) Letting w^A be final wealth if state A happens, and w^C final wealth if state C happens,
 - i. Show that in an interior solution, the optimal x satisfies the following equation. If you cannot get this, assume it anyway in part (c).

$$\frac{r_1^A - r_2^A}{r_2^C - r_1^C} = \frac{u'(w^C)}{u'(w^A)} \quad (*)$$

- ii. Suppose that even at $x = 100$, the RHS of (*) is smaller than the LHS of (*). What should the decision-maker do in this case?
- (c) (6) Under what condition on the return rates does the DM prefer state C to state A, assuming he invests optimally as in (*)?